DENGUE STATISTICS IN MATARA DISTIRCT

Assignment

(SC/2022/12952)

MSF1212: Numerical Analysis I

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Cubic Spline Interpolation

Formula to oblain cubic spline polynomial

```
% Define the x and y coordinates
x = [0,2,4,6,8];
y = [748,1384,1148,545,1773];

% Fit the cubic spline
pp = spline(x, y);

% Extract break points and coefficients
breaks = pp.breaks;
coefs = pp.coefs;

% Display the polynomial equations for each segment
numSegments = size(coefs, 1);
for i = 1:numSegments
    fprintf('Segment %d (from x = %.2f to x = %.2f):\n', i, breaks(i),
breaks(i+1));
    fprintf('y = %.4f * (x - %.2f)^3 + %.4f * (x - %.2f)^2 + %.4f * (x -
%.2f) + %.4f\n', ...
        coefs(i,1), breaks(i), coefs(i,2), breaks(i), coefs(i,3), breaks(i),
coefs(i,4));
    fprintf('\n');
end
```

output

Segment 1 (from x = 0.00 to x = 2.00):

y = 1.7031 * (x - 0.00)^3 + -119.2188 * (x - 0.00)^2 + 549.6250 * (x - 0.00) + 748.0000

Segment 2 (from x = 2.00 to x = 4.00):

y = 1.7031 * (x - 2.00)^3 + -109.0000 * (x - 2.00)^2 + 93.1875 * (x - 2.00) + 1384.0000

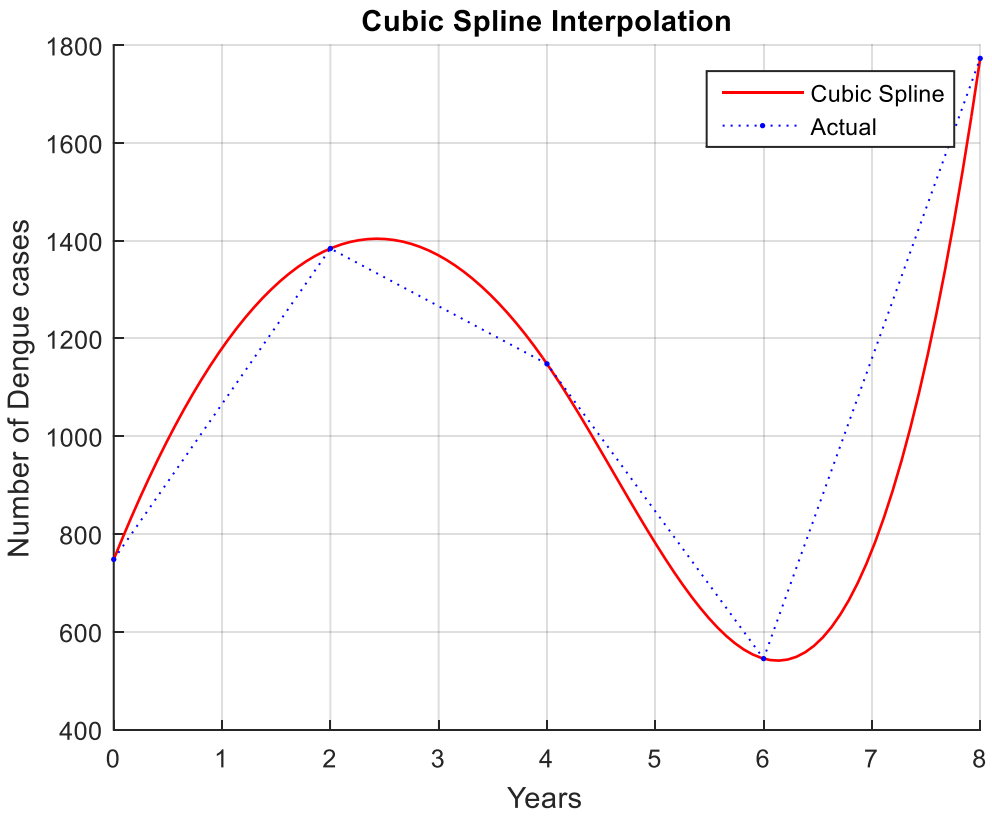
Segment 3 (from x = 4.00 to x = 6.00):

y = 54.6094 * (x - 4.00)^3 + -98.7813 * (x - 4.00)^2 + -322.3750 * (x - 4.00) + 1148.0000

Segment 4 (from x = 6.00 to x = 8.00):

y = 54.6094 * (x - 6.00)^3 + 228.8750 * (x - 6.00)^2 + -62.1875 * (x - 6.00) + 545.0000

Plotting the cubic spline



Cubic Spline MATLAB code

```
x = [0,2,4,6,8];  % Example x-values
y = [748,1384,1148,545,1773];  % Example y-values

% Generate a finer grid for the spline interpolation
x_fine = linspace(min(x), max(x), 100);

% Perform cubic spline interpolation
y_spline = spline(x, y, x_fine);

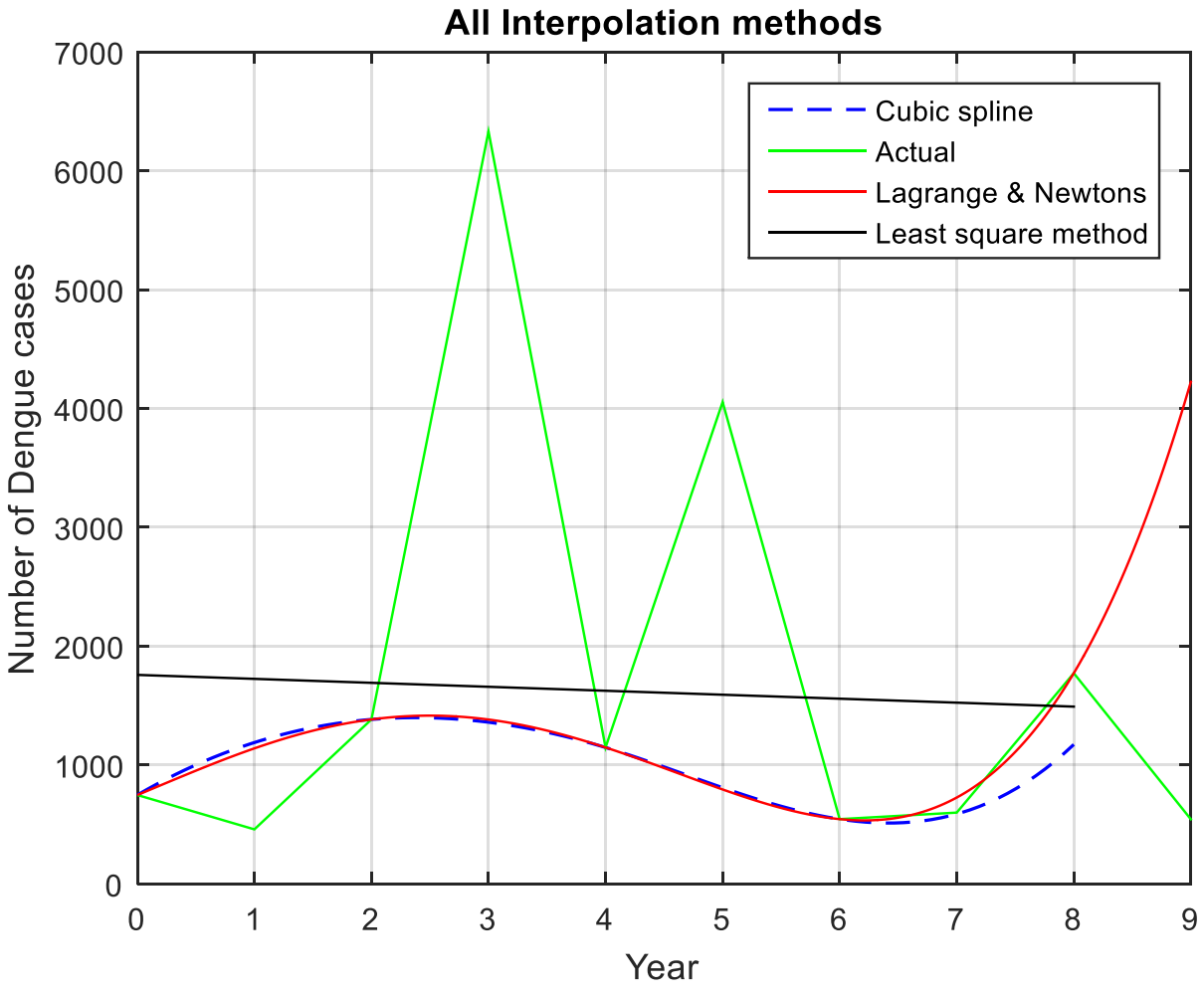
% Plot the original data points
figure;
%plot(x, y, 'o', 'MarkerSize', 8, 'DisplayName', 'Data Points');
hold on;
grid on;

% Plot the cubic spline interpolation
plot(x_fine, y_spline, 'r-', 'LineWidth', 1, 'DisplayName', 'Cubic Spline');
hold on;


plot(x,y,'b.:')
hold on


% Add labels and legend
xlabel('Years');
ylabel('Number of Dengue cases');
title('Cubic Spline Interpolation');
legend('Cubic Spline','Actual');
```

Plotting all the curves in one plot



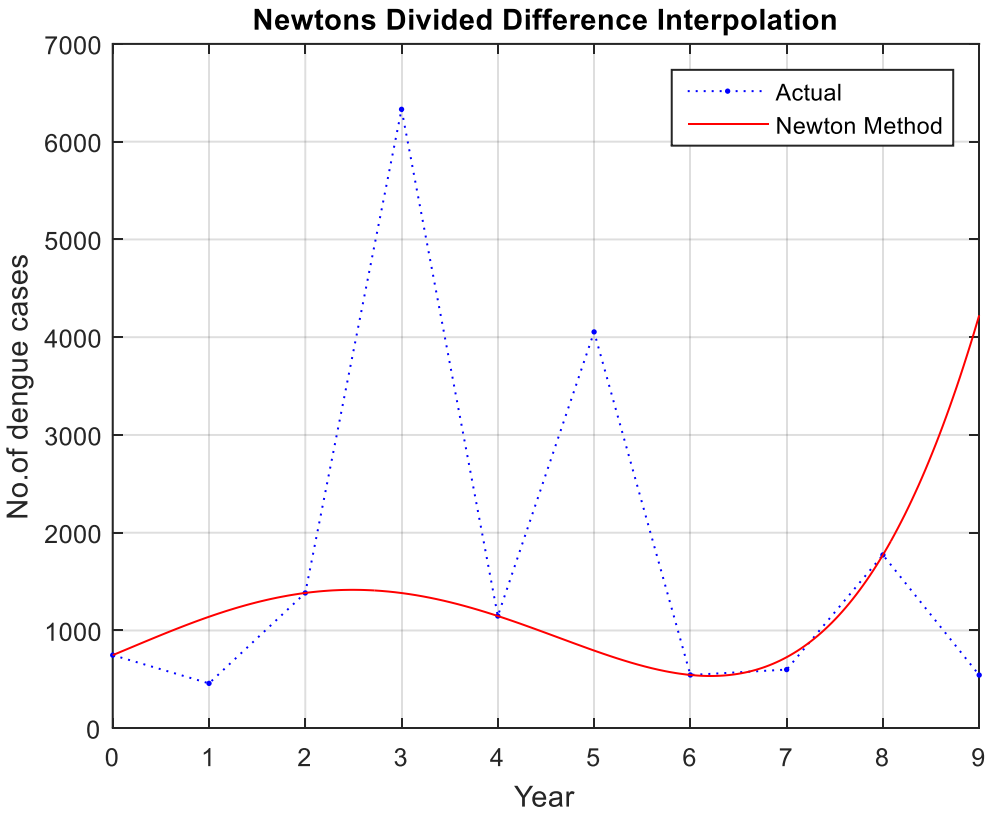
MATLAB code

```
x = [0,2,4,6,8];
y = [748,1384,1148,545,1173];
% Fit the cubic spline
pp = spline(x, y);
% Define a finer set of x values for plotting the spline
x_fine = linspace(min(x), max(x), 100);
% Evaluate the spline at the finer set of x values
y_fine = ppval(pp, x_fine);
% Plot the original data points
figure;

plot(x_fine, y_fine, 'b--', 'LineWidth', 1);
hold on;
x1 = [0,1,2,3,4,5,6,7,8,9];
y1 = [748,459,1384,6331,1148,4054,545,600,1773,544];
plot(x1,y1,'g-')
hold on;
L1=@(x)4.4089*x^4 - 42.3854*x^3 +21.8646*x^2 + 408.5417*x +748;
fplot(L1,[0,9],'r')
hold on;
grid on;
y2=1758.6-33.3636*x
plot(x,y2,'k')
% Add labels and title
xlabel('Year');
ylabel('Number of Dengue cases');
title('All Interpolation methods');
```

```
legend('Cubic spline','Actual','Lagrange & Newtons','Least square method')
```

Newtons Divided Difference Interpolation



MATLAB code

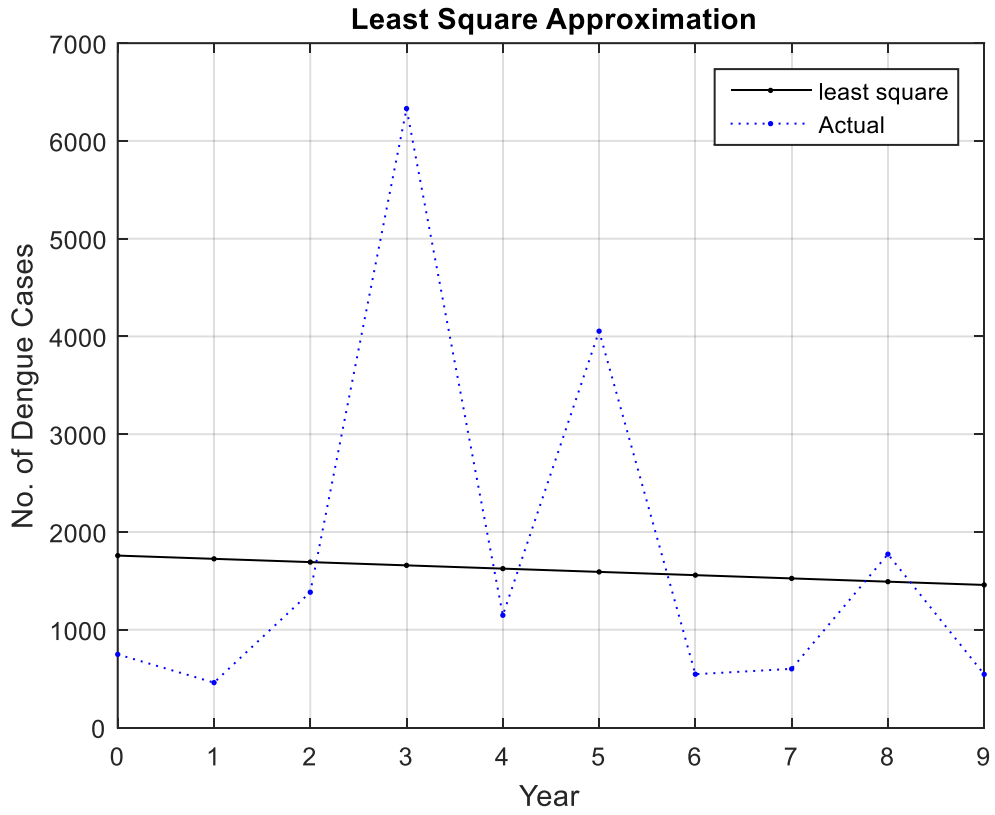
```
x=[0,1,2,3,4,5,6,7,8,9]
y=[748,459,1384,6331,1148,4054,545,600,1773,544]
plot(x,y,'b.:')

hold on
grid on;

N=@(x)4.4089*x^4 - 42.3854*x^3 + 21.8646*x^2 + 408.5417*x + 748;
fplot(N,[0,9],'r')
```

```
xlabel('Year')
ylabel('No.of dengue cases')
title('Newtons Divided Difference Interpolation')
legend('Actual','Newton Method')
```

Least Square Approximation



MATLAB code

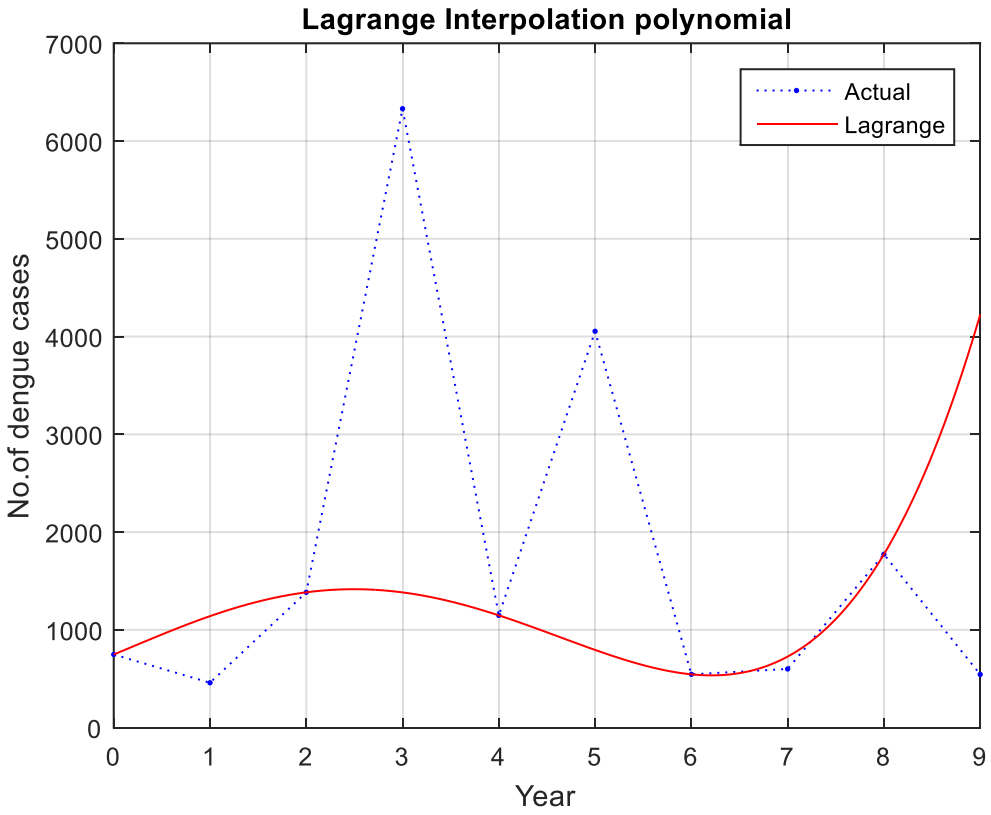
```
y2=1758.6-33.3636*x
plot(x,y2,'k.-')

hold on;
grid on;

x1 = [0,1,2,3,4,5,6,7,8,9]; % example x values
y1 = [748,459,1384,6331,1148,4054,545,600,1773,544]; % example y values
plot(x1,y1,'b.:')
```

```
xlabel('Year');
ylabel('No. of Dengue Cases');
title(' Least Square Approximation');
legend('least square ','Actual');
```

Lagrange Interpolation polynomial



MATLAB code

```
x=[0,1,2,3,4,5,6,7,8,9]
y=[748,459,1384,6331,1148,4054,545,600,1773,544]
plot(x,y,'b.:')

hold on
grid on;

L=@(x)4.4089*x^4 - 42.3854*x^3 + 21.8646*x^2 + 408.5417*x + 748;
```

```
fplot(L,[0,9],'r')

xlabel('Year')
ylabel('No.of dengue cases')
title('Lagrange Interpolation polynomial')
legend('Actual','Lagrange')
```

Lag Missing Values

```
x=[0,2,4,6,8];
y=[748,1384,1148,545,1773];

n=length(x) -1 ;

xp = 3;

sm = 0;

for i= 1: n+1
    pr= 1;
    for j =1:n+1
        if j~=i

            pr =pr*(xp -x(j))/(x(i) -x(j));
        end
    end
    sm=sm + y(i)*pr;
end
yp =sm
```

Lagrange polynomial

```
syms x %x1 x2 x3 y1 y2 y3

xx=[0 2 4 6 8];
y=[748 1384 1148 545 1773];
L2=[];
for i=1:length(xx)
 LL=[];
for j=1:length(xx)
   if i~=j
  L= (x-xx(j))/(xx(i)-xx(j));
  LL=[LL,L];
  L1=prod(LL)*y(i);

   end
end
L2=[L2,L1];
end
L2;
```

```
pol=simplify(sum(L2))
vpa(pol)
```

Lag plot

```
x=[2,5,8,11,14];
y=[2837,1729,3130,4036,3022];

plot(x,y,'r')

hold on

L1=@(x)10.586*x+2787.2;
fplot(L1,[2,14],'b')

hold on

L2=@(x)0.8122*x^4-39.66*x^3+605.17*x^2-3023.58*x+7167.79;
fplot(L2,[2,14],'g')

xlabel('year')
ylabel('no.of dengue cases')
title('Lagrange Interpolation')
legend('Actual','Approximation 1','Approximation 2')
```

Cubic spline

```
% Define the x and y coordinates
x = [0,2,4,6,8]; % example x values
y = [748,1384,1148,545,1773]; % example y values

% Fit the cubic spline
pp = spline(x, y);

% Extract break points and coefficients
breaks = pp.breaks;
coefs = pp.coefs;

% Display the polynomial equations for each segment
numSegments = size(coefs, 1);
for i = 1:numSegments
    fprintf('Segment %d (from x = %.2f to x = %.2f):\n', i, breaks(i),
breaks(i+1));
```

```
    fprintf('y = %.4f * (x - %.2f)^3 + %.4f * (x - %.2f)^2 + %.4f * (x -
%.2f) + %.4f\n', ...
        coefs(i,1), breaks(i), coefs(i,2), breaks(i), coefs(i,3), breaks(i),
coefs(i,4));
    fprintf('\n');
end
```

Cubic plot

```
% Define the x and y coordinates
x = [0,2,4,6,8]; % example x values
y = [748,1384,1148,545,1773]; % example y values

% Fit the cubic spline
pp = spline(x, y);

% Define a finer set of x values for plotting the spline
x_fine = linspace(min(x), max(x), 100);

% Evaluate the spline at the finer set of x values
y_fine = ppval(pp, x_fine);

% Plot the original data points
figure;
plot(x, y, 'o', 'MarkerSize', 8, 'MarkerFaceColor', 'r');
hold on;

% Plot the cubic spline
plot(x_fine, y_fine, 'b-', 'LineWidth', 1.5);

%hold on;
%x1 = [0,1,2,3,4,5,6,7,8,9]; % example x values
%y1 = [748,459,1384,6331,1148,4054,545,600,1773,544]; % example y values

%plot(x1,y1,'g')

% Add labels and title
xlabel('x');
ylabel('y');
title('Cubic Spline Interpolation');
```

combine plot

```
% Define the x and y coordinates
x = [1,3,5,7,9]; % example x values
y = [223,3244,333,170,137]; % example y values

% Fit the cubic spline
```

```
pp = spline(x, y);

% Define a finer set of x values for plotting the spline
x_fine = linspace(min(x), max(x), 100);

% Evaluate the spline at the finer set of x values
y_fine = ppval(pp, x_fine);

% Plot the original data points
figure;
plot(x, y, 'o', 'MarkerSize', 11, 'MarkerFaceColor', 'r');
hold on;

% Plot the cubic spline
plot(x_fine, y_fine, 'b-', 'LineWidth', 1.5);

hold on;
x1 = [0,2,4,6,8]; % example x values
y1 = [313,475,854,0,540]; % example y values

plot(x1,y1,'g')

hold on;

L1=@(x)-24.4219*x^4 + 651.5833*x^3 -4899.2813*x^2 +13813.9167*x -9313.7969;
fplot(L1,[0,9],'r')

hold on;
y2=628.9-33.7667*x
plot(x,y2,'k')

% Add labels and title
xlabel('x');
ylabel('y');
title('Cubic Spline Interpolation');
```